Causal identification in macro II

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Local projection

- Suppose we actually manage to directly observe the shock of interest, say, the first one ε_1.
- Example: We use historical documents to measure all surprise changes to tax law in the U.S. (ε_1: tax shock). "Narrative approach".
- We still normalize $\text{Var}(\varepsilon_{1,t}) = 1$.
- Then makes sense to identify impulse responses by direct projection:

$$\Theta_{i1,\ell} = E^*(y_{i,t+\ell} \mid \varepsilon_{1,t} = 1) - E^*(y_{i,t+\ell} \mid \varepsilon_{1,t} = 0), \quad \ell = 0, 1, 2, \ldots$$

- These direct projections are called **local projections (LP)** (Jordà, 2005). They are just multi-step forecasts, but we care about the predictive power of only one variable (the shock).

Local projection: estimation

$$\Theta_{i1,\ell} = E^*(y_{i,t+\ell} \mid \varepsilon_{1,t} = 1) - E^*(y_{i,t+\ell} \mid \varepsilon_{1,t} = 0), \quad \ell = 0, 1, 2, \ldots$$

- Can estimate by a direct linear regression, *separately* for each horizon $\ell = 0, 1, 2, \ldots$:

$$y_{i,t+\ell} = \hat{\mu}_\ell + \hat{\beta}_\ell \varepsilon_{1,t} + \text{controls}_t + \hat{e}_{t+\ell|t}.$$

- Estimated IRF: $(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \ldots)$
- Control variables should be orthogonal to $\varepsilon_{1,t}$ (e.g., lagged var's). Controls not needed for consistency, but may improve efficiency.
- For any horizon ℓ, the population projection error $e_{t+\ell|t}$ could be serially correlated. Suggests HAC standard errors (more soon).

Application: Effect of monetary shock

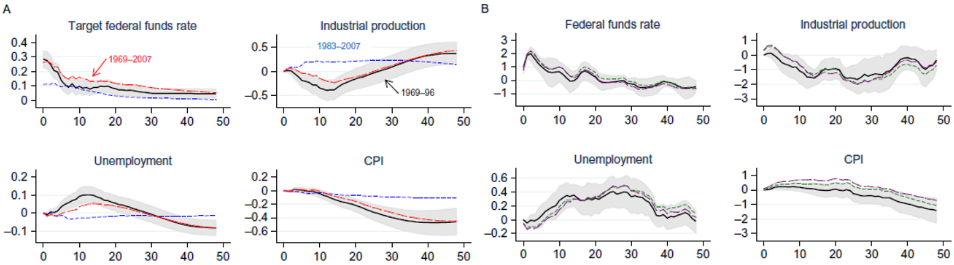


Fig. 2 Romer and Romer monetary shock. (A) Coibion VAR 1969m3–1996m12: *solid black lines*; 1983m1–2007m12: *short dashed blue* (*dark gray* in the print version) *lines*; 1969m3–2007m12: *long-dashed red* (*gray* in the print version) *lines*. (B) Jordà local projection method, 1969m3–1996m12 recursiveness assumption: *solid black lines*; no recursiveness assumption: *short dashed green* (*dark gray* in the print version) *lines*; no recursiveness assumption, FAVAR controls: *long dashed purple* (*dark gray* in the print version) *lines*. Light gray bands are 90% confidence bands.

Source: Ramey (2016) handbook chapter

Distributed lag regression

- Since $\varepsilon_{1,t} \sim WN$, we could instead run a single **distributed lag** regression
$$y_{i,t} = \tilde{\mu} + \sum_{\ell=0}^{K} \tilde{\beta}_\ell \varepsilon_{1,t-\ell} + \tilde{e}_t.$$
- Estimated IRF: $(\tilde{\beta}_0, \tilde{\beta}_1, \ldots, \tilde{\beta}_K)$.
- It's harder to use controls here, so not as popular as local projections.

LP vs. VAR

- How do LP and VAR impulse response estimands differ?
- Consider a LP of $y_{i,t}$ on a variable x_t (which need not be a "shock"), and where we flexibly control for lagged data:

$$y_{i,t+\ell} = \beta_\ell x_t + \sum_{h=1}^{\infty} \gamma'_{\ell,h} y_{t-h} + e_{t+\ell|t}. \tag{†}$$

(Read this as a population linear projection.)

- x_t is the first element of y_t. Assume everything mean zero.
- By Frisch-Waugh Theorem (residual regression), can obtain same β_ℓ as in (†) from the regression

$$y_{i,t+\ell} = \beta_\ell \eta_{1,t} + \tilde{e}_{t+\ell|t},$$

where we define the Wold residuals $\eta_t \equiv y_t - E^*(y_t \mid \{y_\tau\}_{\tau<t})$.

$$\implies \beta_\ell = \frac{E(y_{i,t+\ell}\eta_{1,t})}{E(\eta_{1,t}^2)}.$$

LP vs. VAR (cont.)

- Now consider the VAR(∞) representation of the data:

$$A(L)y_t = \eta_t, \quad A(L) = I_n - \textstyle\sum_{\ell=1}^{\infty} A_\ell L^\ell.$$

(Again, read this as a population linear projection.)

- Invert to get the reduced-form MA representation

$$y_t = B(L)\eta_t, \quad B(L) = I_n + \textstyle\sum_{\ell=1}^{\infty} B_\ell L^\ell \equiv A(L)^{-1}.$$

- To simplify the math, let's assume $E(\eta_{1,t}\eta_{j,t}) = 0$ for $j > 1$. (We don't need this for the general result stated on slide 9.) Then

$$E(y_{i,t}\eta_{1,t-\ell}) = B_{i,1,\ell}E(\eta_{1,t-\ell}^2).$$

- Thus, the VAR-implied reduced-form impulse response of $y_{i,t}$ wrt. the first innovation at horizon ℓ is given by

$$B_{i,1,\ell} = \frac{E(y_{i,t+\ell}\eta_{1,t})}{E(\eta_{1,t}^2)} = \beta_\ell \quad \text{(cf. previous slide)}.$$

LP vs. VAR (cont.)

- Conclusion: VAR and LP impulse response estimands are *the same*!
- Intuition:
  - VAR(p) impulse response = mean-square-optimal forecast given second moments implied by VAR(p) model.
  - As $p \to \infty$, VAR captures all second moments in the actual DGP.
  - Thus, VAR(∞) impulse response = mean-square optimal forecast given actual second moments = LP.
- Note: This is a *nonparametric* result. Doesn't assume a particular SVMA model or SVAR(p) model, or that x_t is a "shock".
- The previous discussion only considered reduced-form VAR IRs, and only considered lags as controls in the LP. We will now generalize.

LP vs. VAR: control variables

- Consider more general LP with contemporaneous controls r_t:

$$y_{i,t+\ell} = \beta_\ell x_t + \delta_\ell' r_t + \sum_{h=1}^{\infty} \gamma_{\ell,h}' y_{t-h} + e_{t+\ell|t}.$$

- y_t contains both x_t and r_t (and possibly other series).
- Plagborg-Moller & Wolf (2021) prove that β_ℓ nonparametrically equals the horizon-ℓ impulse response from a recursively identified SVAR(∞) where:
  - r_t is ordered first.
  - Then x_t.
  - Then the other variables.
- Lesson: Recursive SVARs are the same as LPs with particular choice of control variables, and vice versa.
- Note: For the LP and SVAR IRFs to coincide, we need to scale the LP IRF so it corresponds to an innovation of variance 1 (like SVAR).

LP vs. VAR: finite lag length

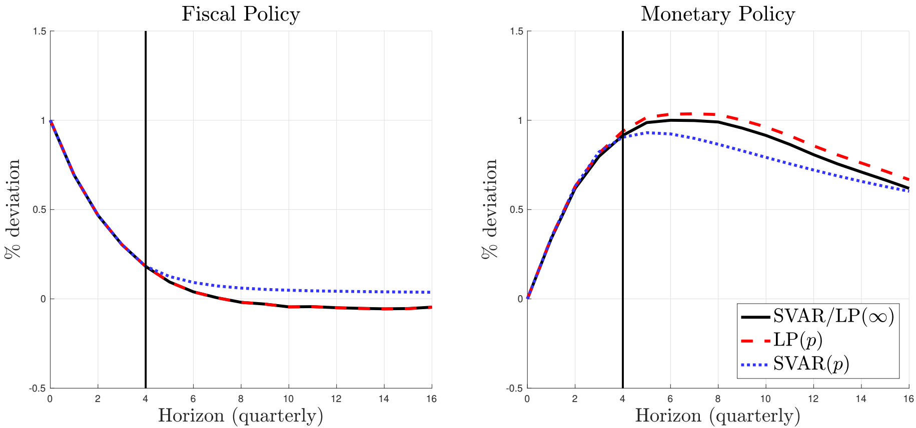
- The LP/VAR equivalence result relies on infinite lag length $p = \infty$.
- What if p is finite in the projection and the SVAR?

$$y_{i,t+\ell} = \beta_\ell^{(p)} x_t + \sum_{h=1}^{p} (\gamma_{\ell,h}^{(p)})' y_{t-h} + e_{t+\ell|t}^{(p)},$$

$$A^{(p)}(L) y_t = \eta_t^{(p)}, \quad A^{(p)}(L) = I_n - \sum_{\ell=1}^{p} A_\ell^{(p)} L^\ell.$$

- Plagborg-Moller & Wolf (2021) show:
  - If x_t is unpredictable, then SVAR(p) and LP(p) impulse responses coincide for horizons $\ell = 0, 1, 2, \ldots, p$. They generally differ for $\ell > p$.
  - If x_t is not unpredictable, SVAR(p) and LP(p) impulse responses generally differ at all horizons. But difference is typically negligible for small ℓ.

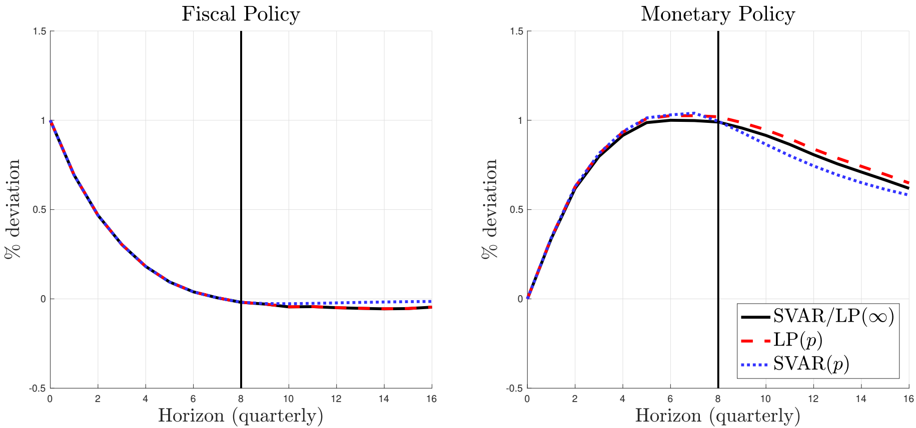
LP vs. VAR: illustration



$p = 4$

Source: Plagborg-Moller & Wolf (2021)

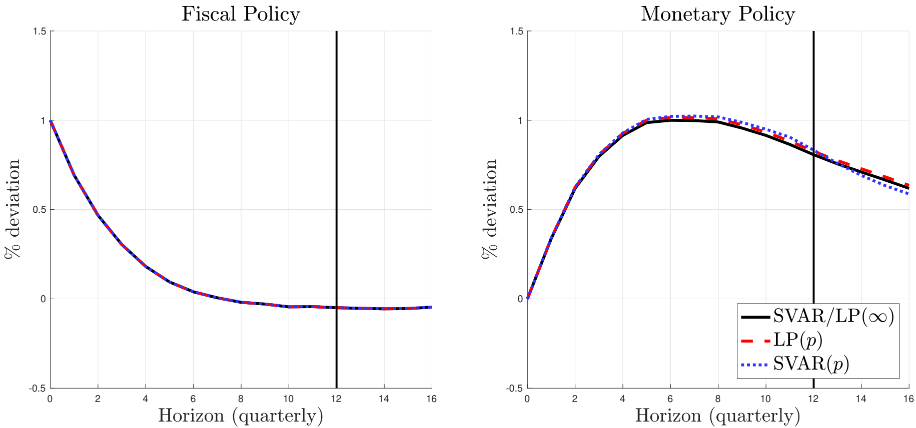
LP vs. VAR: illustration



$p = 8$

Source: Plagborg-Moller & Wolf (2021)

LP vs. VAR: illustration



$p = 12$

Source: Plagborg-Moller & Wolf (2021)